

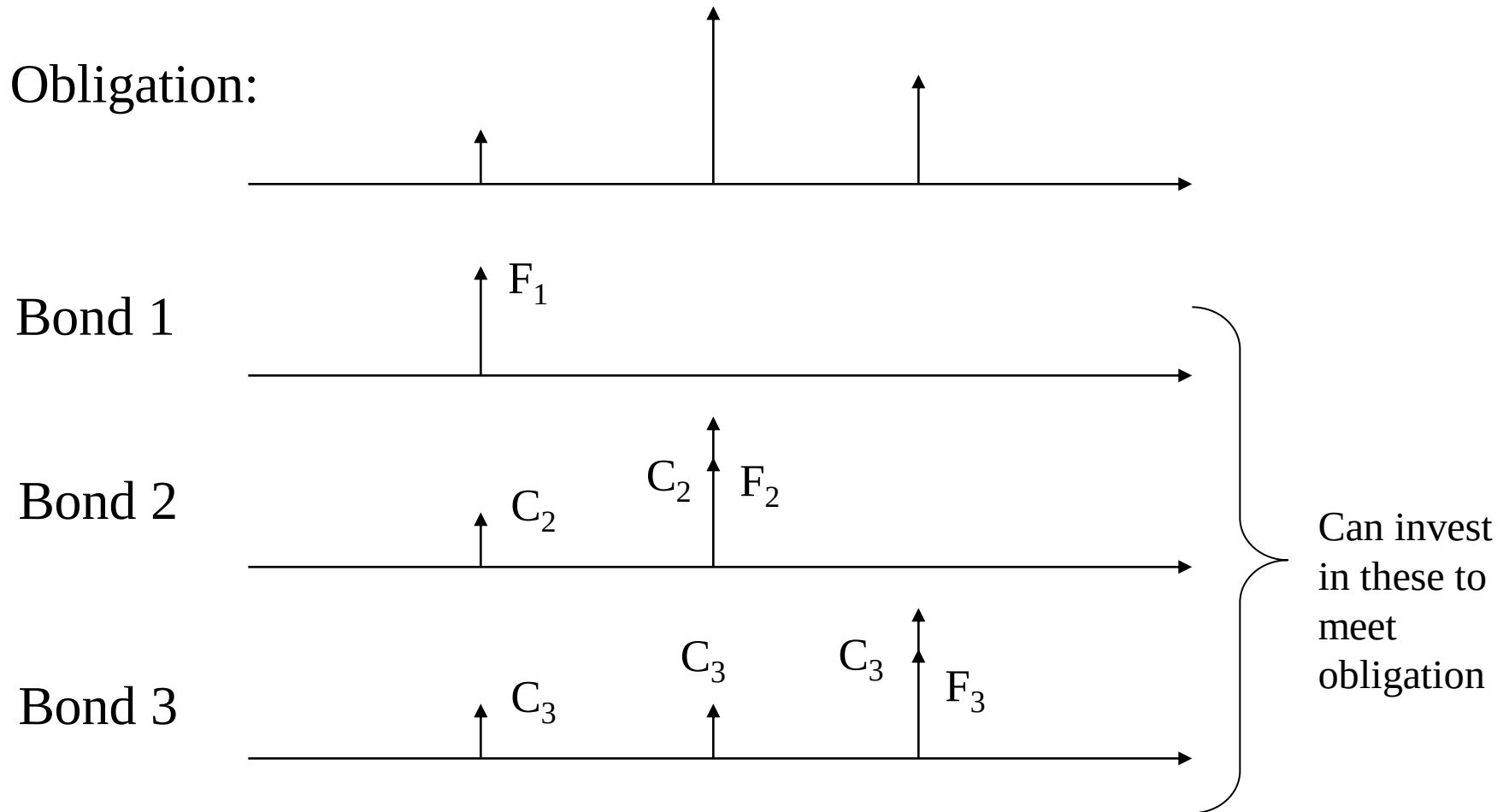
Linear Programming Example: Cash Matching Problem

A series of future obligations must be met.

You have various bonds you can invest in.

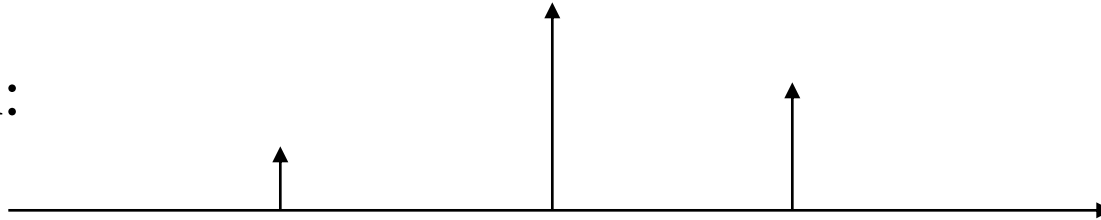
You would like to construct a portfolio to meet the obligations at minimum cost.

Cash Matching in Pictures



Cash Matching in Pictures

Obligation:



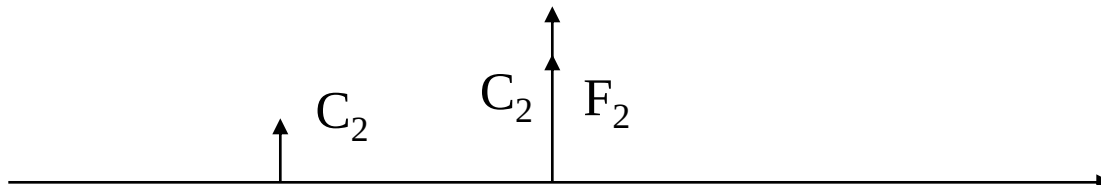
Bond 1



of bonds

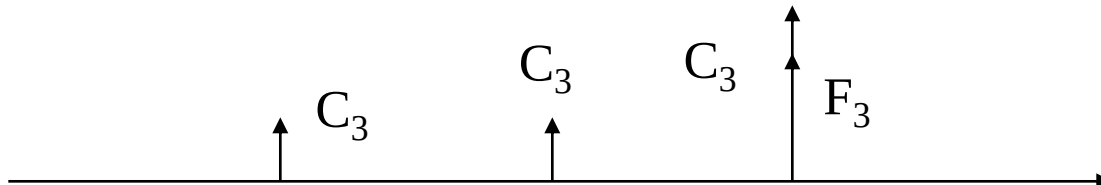
x_1

Bond 2



x_2

Bond 3



x_3

Choose portfolio of minimum cost that has enough money to meet the obligation.

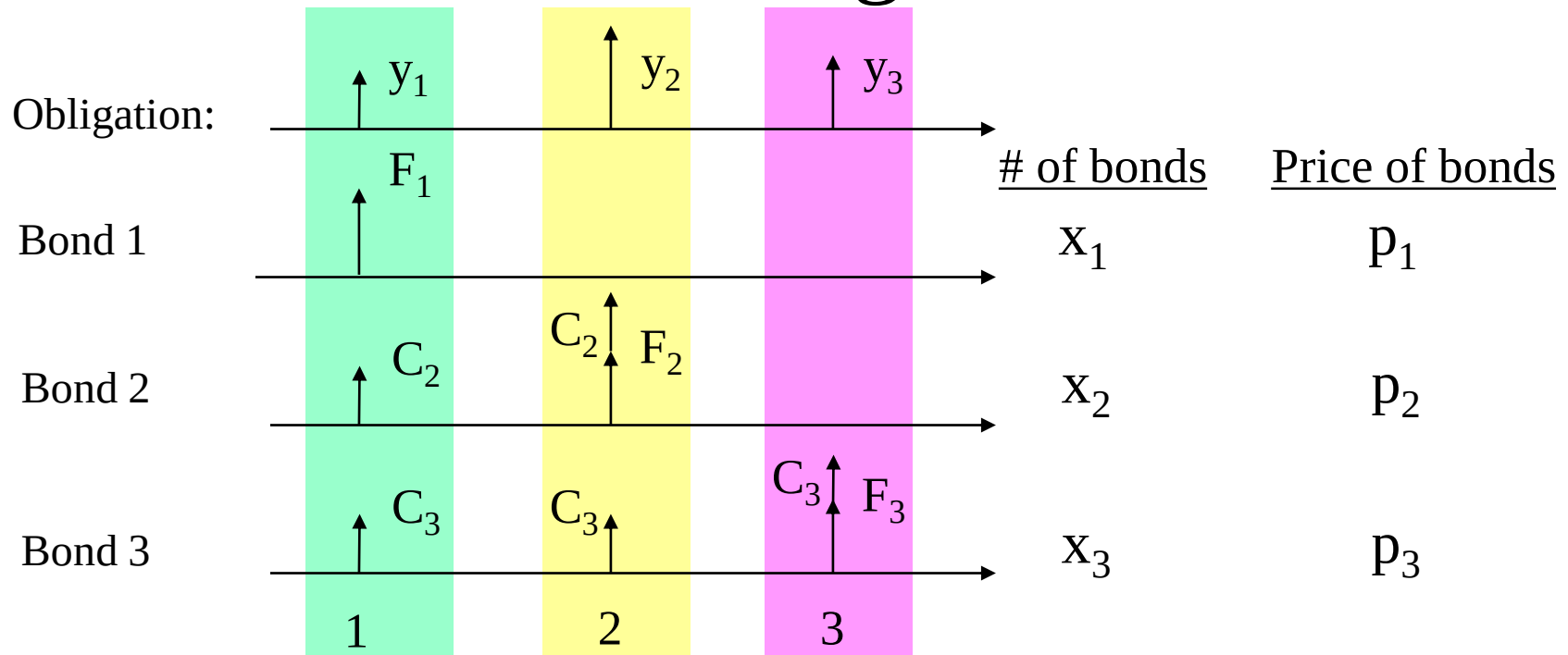
Cash Matching Assumptions

Can solve the cash matching problem under different assumptions:

- 1) No reinvestment of excess cash is allowed.
- 2) Excess cash can be held at zero interest rate from one period to the next.
- 3) Excess cash can be reinvested at the appropriate forward rate.



Cash Matching in Pictures



Without reinvestment

$$\min_{x_1, x_2, x_3} p_1 x_1 + p_2 x_2 + p_3 x_3$$

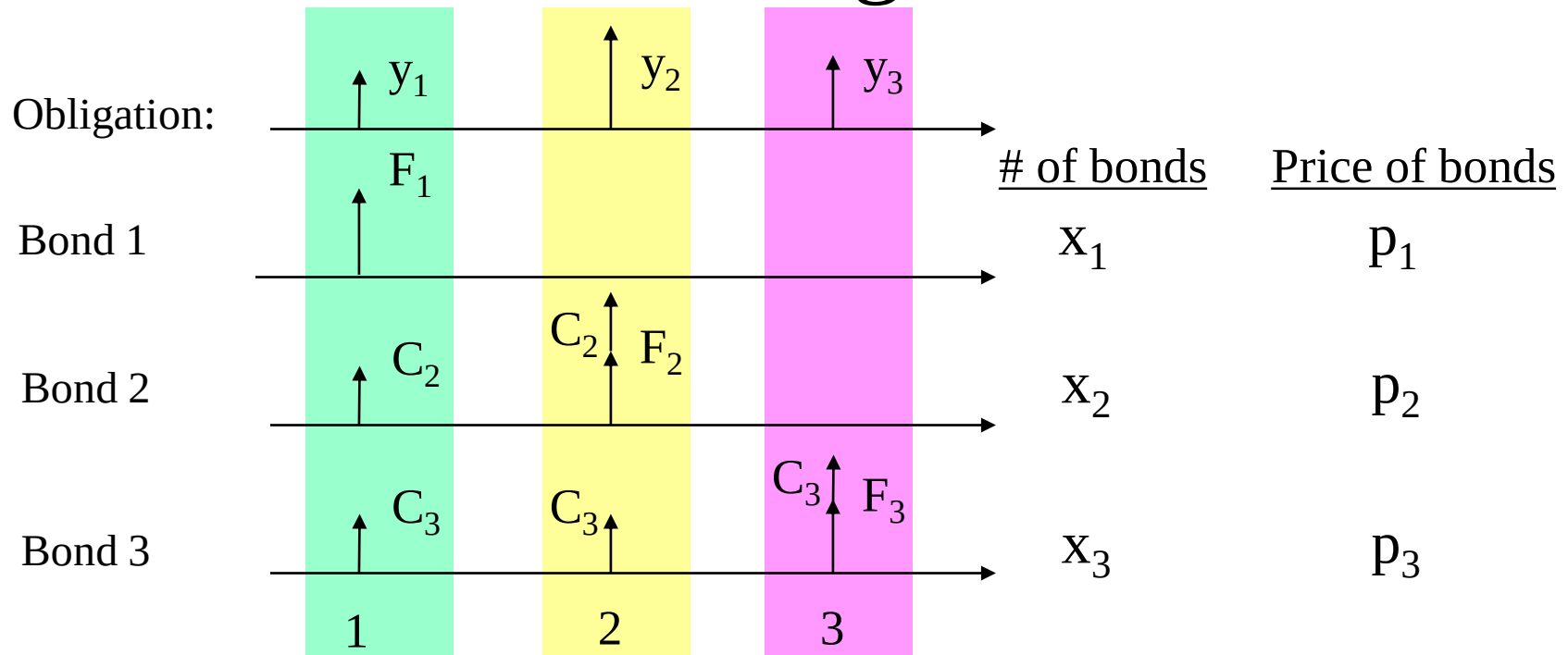
$$1 \quad F_1 x_1 + C_2 x_2 + C_3 x_3 \geq y_1$$

$$2 \quad (F_2 + C_2) x_2 + C_3 x_3 \geq y_2$$

$$3 \quad (F_3 + C_3) x_3 \geq y_3$$



Cash Matching in Pictures



With reinvestment at zero rate

$$\min_{x_1, x_2, x_3} p_1 x_1 + p_2 x_2 + p_3 x_3$$

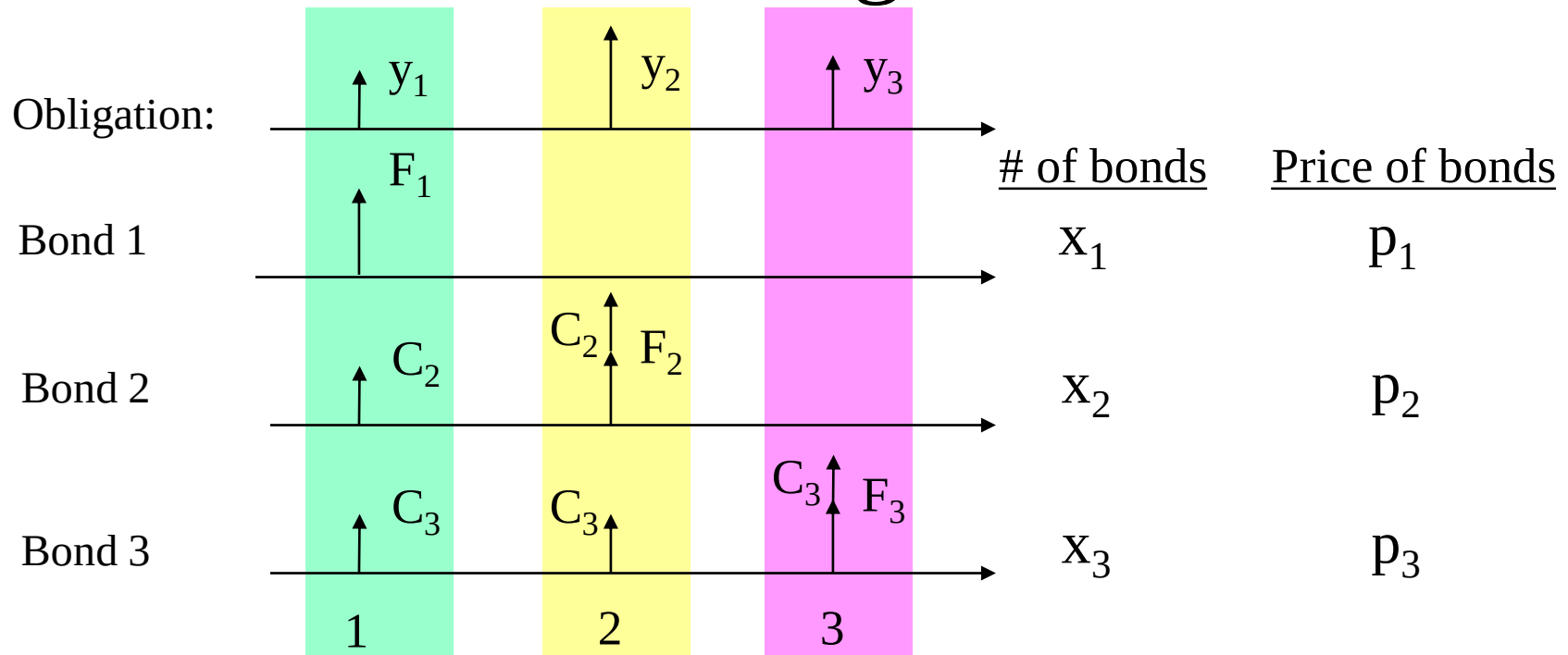
$$1 \quad F_1 x_1 + C_2 x_2 + C_3 x_3 - z_1 \geq y_1 \quad z_1 \geq 0$$

$$2 \quad (F_2 + C_2) x_2 + C_3 x_3 + z_1 - z_2 \geq y_2 \quad z_2 \geq 0$$

$$3 \quad (F_3 + C_3) x_3 + z_2 \geq y_3$$



Cash Matching in Pictures



With reinvestment at forward rate

$$\min_{x_1, x_2, x_3} p_1 x_1 + p_2 x_2 + p_3 x_3$$

$$1 \quad F_1 x_1 + C_2 x_2 + C_3 x_3 - z_1 \geq y_1 \quad z_1 \geq 0$$

$$2 \quad (F_2 + C_2) x_2 + C_3 x_3 + (1 + f_{1,2}) z_1 - z_2 \geq y_2$$

$$3 \quad (F_3 + C_3) x_3 + (1 + f_{2,3}) z_2 \geq y_3 \quad z_2 \geq 0$$

Cash Matching									
			Bonds	1	2	3	4		
			Coupon	3%	4%	2%	5%		
No Reinvestment									
Term Structure				Bonds					
s(0,i)	f(i-1,i)	d(0,i)	Time	1	2	3	4	Amount	Obligation
3%	3.0%	0.970446	1	103	4	2	5	10.502406	>= 10
3.50%	4.0%	0.932394	2		104	2	5	200	>= 200
4%	5.0%	0.88692	3			102	5	30	>= 30
4.20%	4.8%	0.845354	4				105	50	>= 50
			Price	99.95589	100.8507	94.27156	102.711		
			Portfolio	0	1.894976	0.270775	0.47619		
			Amount in	0	191.1097	25.52638	48.90998	265.54608	
								Total Price	

Reinvest at Zero Interest													
Term Structure				Bonds									
s(0,i)	f(i,i-1)	d(0,i)	Time	1	2	3	4	Reinvest at zero interest			Amount		Obligation
3%	3.0%	0.970446	1	103	4	2	5	-1			10	>=	10
3.50%	4.0%	0.932394	2		104	2	5	1	-1		200	>=	200
4%	5.0%	0.88692	3			102	5		1	-1	30	>=	30
4.20%	4.8%	0.845354	4				105			1	50	>=	50
			Price	99.95589	100.8507	94.27156	102.711	0	0	0			
			Portfolio	0	1.890324	0.270775	0.47619	0.4837985	0	0			
			Amount in	0	190.6406	25.52638	48.90998	0	0	0	265.07693		
											Total Price		

Term Structure				Bonds									
s(0,i)	f(i,i-1)	d(0,i)	Time	1	2	3	4	Reinvest at zero interest			Amount		Obligation
3%	3.0%	0.970446	1	103	4	2	5	-1			10	>=	10
3.50%	4.0%	0.932394	2		104	2	5	1-1			200	>=	200
4%	5.0%	0.88692	3			102	5	1-1			30	>=	30
4.20%	4.8%	0.845354	4				105	1			50	>=	50
			Price	99.95589	100.8507	94.27156	102.711	000					
			Portfolio	0	1.890324	0.270775	0.47619	0.483798500					
			Amount in	0	190.6406	25.52638	48.90998	000			265.07693		
											Total Price		

Term Structure				Bonds				Reinvest at forward rate			Amount		Obligation
s(0,i)	f(i,i-1)	d(0,i)	Time	1	2	3	4						
3%	3.000%	0.970446	1	103	4	2	5	-1			10	>=	10
3.50%	4.000%	0.932394	2		104	2	5	1.0408108	-1		200	>=	200
4%	5.000%	0.88692	3			102	5		1.05127	-1	30	>=	30
4.20%	4.800%	0.845354	4				105			1.049171	50	>=	50
			Price	99.95589	100.8507	94.27156	102.711	0	0	0			
			Portfolio	0.21037202	1.782777	0.262646	0.384342	21.24643	9.96928	9.192063			
			Amount in	21.0279227	179.7943	24.76009	39.47618	0	0	0	265.05852		
											Total Price		

Term Structure				Bonds									
s(0,i)	f(i,i-1)	d(0,i)	Time	1	2	3	4	Reinvest at forward rate			Amount		Obligation
3%	3.000%	0.970446	1	103	4	2	5	-1			10	>=	10
3.50%	4.000%	0.932394	2		104	2	5	1.0408108	-1		200	>=	200
4%	5.000%	0.88692	3			102	5		1.05127	-1	30	>=	30
4.20%	4.800%	0.845354	4				105			1.049171	50	>=	50
			Price	99.95589	100.8507	94.27156	102.711	0	0	0			
			Portfolio	0.21037202	1.782777	0.262646	0.384342	21.24643	9.96928	9.192063			
			Amount in	21.0279227	179.7943	24.76009	39.47618	0	0	0	265.05852		
											Total Price		